Effects of a high-fat meal on pulmonary function in healthy subjects.

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Obesity has important health consequences, including elevating risk for heart disease, diabetes, and cancer. A high-fat diet is known to contribute to obesity. Little is known regarding the effect of a high-fat diet on pulmonary function, despite the dramatic increase in the prevalence of respiratory ailments (e.g., asthma). The purpose of our study was to determine whether a high-fat meal (HFM) would increase airway inflammation and decrease pulmonary function in healthy subjects. Pulmonary function tests (PFT) (forced expiratory volume in 1-s, forced vital capacity, forced expiratory flow at 25-75% of vital capacity) and exhaled nitric oxide (eNO; airway inflammation) were performed in 20 healthy (10 men, 10 women), inactive subjects (age 21.9 +/- 0.4 years) pre and 2 h post HFM (1 g fat/1 kg body weight; 74.2 +/- 4.1 g fat). Total cholesterol, triglycerides, and C-reactive protein (CRP; systemic inflammation) were determined via a venous blood sample pre and post HFM. Body composition was measured via dual energy X-ray absorptiometry. The HFM significantly increased total cholesterol by 4 +/- 1%, and triglycerides by 93 +/- 3%. ENO also increased ($p < 0.05$) due to the HFM by 19 +/- 1% (pre 17.2 +/- 1.6; post 20.6 +/- 1.7 ppb). ENO and triglycerides were significantly related at baseline and post-HFM ($r = 0.82$, 0.72 respectively). Despite the increased eNO, PFT or CRP did not change ($p > 0.05$) with the HFM. These results demonstrate that a HFM, which leads to significant increases in total cholesterol, and especially triglycerides, increases exhaled NO. This suggests that a high-fat diet may contribute to chronic inflammatory diseases of the airway and lung.